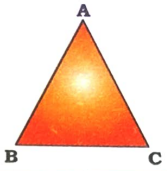
TRIANGLE

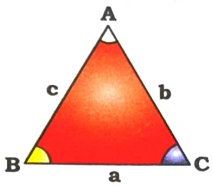
A triangle is a polygon with three sides and three angles. It is the fundamental shape in geometry formed by connecting three non-collinear points.



Fundamental Properties of Triangle

☞ Sum of all three angles of a triangle is always 180°

$\angle A + \angle B + \angle C = 180°$



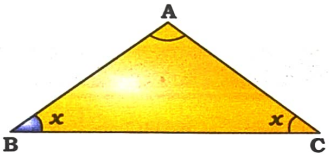
Ex. One of the angles of a triangle is 108° and the other two angles are equal. What is the measure of each of these equal angles?

HINTS We know,

$\angle A + \angle B + \angle C = 180°$

$\Rightarrow 108° + x + x = 180°$

$\Rightarrow 2x = 72° \Rightarrow x = 36°$



☞ (i) Angles opposite to the equal sides of a triangle are equal.

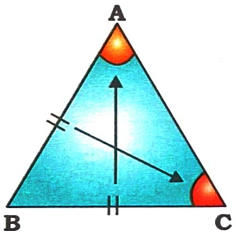
(ii) Sides opposite to the equal angles of a triangle are equal.

In ΔABC, If AB = BC, then

$\angle A = \angle C$

In ΔABC, If $\angle A = \angle C$, then

AB = BC

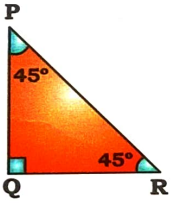


Ex. In ΔPQR, $\angle Q = 90°$, PQ = 8 cm and $\angle PRQ = 45°$. Find the length of QR.

HINTS

Using property 2(ii),

QR = PQ = 8 cm



☞ (i) The angle opposite to the greater side is always greater than the angle opposite to the smaller side.

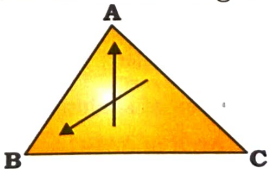
(ii) The side opposite to the greater angle is always greater than the side opposite to the smaller angle.

In ΔABC, If BC > AC, then

$\angle A > \angle B$

In ΔABC, If $\angle A > \angle C$, then

BC > AB



Ex. The ratio of the angles, ∠P, ∠Q and ∠R of a ΔPQR is 2 : 4 : 9, then which of the following is true?

(a) PQ > QR > RP

(b) PQ > RP > QR

(c) QR > RP > PQ

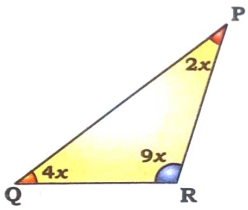
(d) PR > PQ > QR

HINTS

Here, In ΔPQR

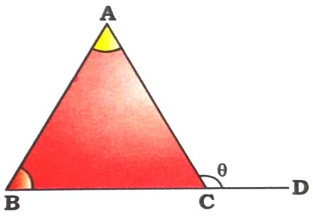
$9x > 4x > 2x$

So, PQ > PR > QR



☞ If a side of triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.

$\angle ACD = \angle CAB + \angle ABC$



Ex. The side BC of ΔABC is produced to D. If $\angle ACD = 112°$ and $\angle B = \frac{3}{4}\angle A$, then find the measure of ∠B.

HINTS We know,

Exterior angle (∠ACD)

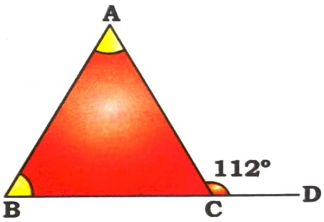
$= \angle A + \angle B$

$\because \angle B = \frac{3}{4}\angle A$

$\Rightarrow \angle A = \frac{4}{3}\angle B$ ....(i)

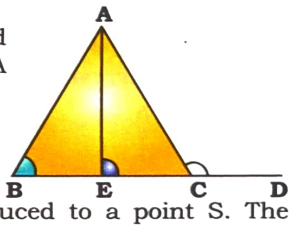
$\therefore 112° = \frac{4}{3}\angle B + \angle B$

$\Rightarrow \angle B = \frac{3}{7} \times 112° = 48°$



☞ In ΔABC, the side BC is produced to D and angle bisector of ∠A meets BC at E, then

$\angle ABC + \angle ACD = 2\angle AEC$.



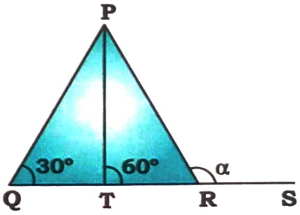
Ex. The side QR of ΔPQR is produced to a point S. The bisector of ∠P meets side QR at T. If $\angle PQR = 30°$ and $\angle PTR = 60°$, find ∠PRS.

HINTS

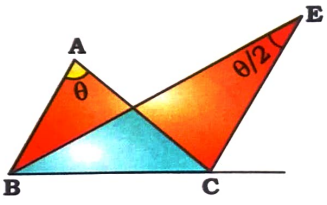
$\angle PQR + \angle PRS = 2 \times \angle PTR$

$\Rightarrow 30° + \alpha = 2 \times 60°$

$\Rightarrow \alpha = 120° - 30° = 90°$



☞ In a triangle, the angle formed between internal bisector of one base angle and external bisector of the other base angle is half of the remaining vertex angle.



According to this property, $\angle BEC = \frac{\angle A}{2}$.

Ex. In ΔABC, ∠B = 72° and ∠C = 44°. Side BC is produced to D. Then bisectors of ∠B and ∠ACD meet at E. What is the measure of ∠BEC?

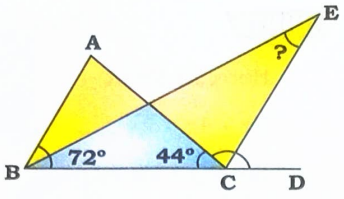
HINTS We know,

$\angle A + \angle B + \angle C = 180°$

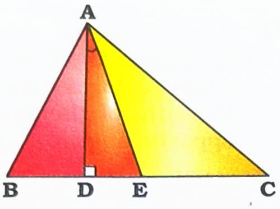
$\Rightarrow \angle A = 180° - 72° - 44°$

$\Rightarrow \angle A = 64°$

$\therefore BEC = \frac{\angle A}{2} = \frac{64°}{2} = 32°$



☞ The angle between perpendicular drawn from a vertex to opposite side and angle bisector of that vertex angle is half of difference between other two remaining vertex angles.



If AD ⊥ BC and AE is angle bisector of ∠A, then

$\angle DAE = \left|\frac{\angle B - \angle C}{2}\right|$

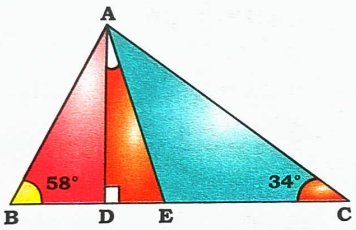
Ex. In ΔABC, AD is perpendicular to BC and AE is the bisector of ∠BAC. If ∠ABC = 58° and ∠ACB = 34°, then find the measure of ∠DAE.

HINTS

$\angle DAE = \frac{\angle B - \angle C}{2}$

$= \frac{58° - 34°}{2} = \frac{24°}{2}$

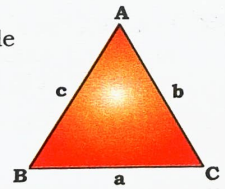
$= 12°$



☞ (i) Sum of any two sides of a triangle is always greater than the third side.

$a + b > c \quad , \quad b + c > a$

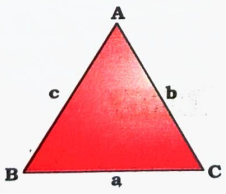
$c + a > b$



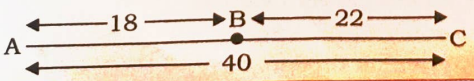
(ii) The difference of the length of any two sides of a triangle is always smaller than the third side.

$|a - b| < c \quad , \quad |b - c| < a$

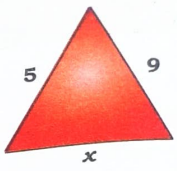
$|c - a| < b$



1. If one side of a triangle is greater than the sum of the other two sides then a triangle can't be formed.
2. When the length of one side is equal to sum of the length of other two sides then the points are collinear & triangle is not formed (just a straight line back and forth).i.e., if a + b = c then point A, B and C are collinear.



Ex. Three sides of a triangle are 5 cm, 9 cm, and x cm. The minimum integral value of x is.



HINTS

Value of x lies between, $4 < x < 14$

Thus, Minimum integral value of x is 5.

Ex. If the sides of a triangle are 7, 12 and x, and x is an integer, then find the number of possible values of x.

HINTS $12 + 7 > x > 12 - 7$

$\Rightarrow 19 > x > 5$

$x = (19 - 5) - 1 = 13$

Alternatively

No. of possible values of x = 2 × smallest side − 1

= 2 × 7 − 1 = 13

"Number of Possible value of x = 2 × smallest side − 1"

Area of Triangle

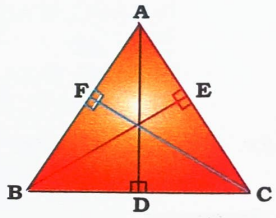
(A) Area of ΔABC = $\frac{1}{2}$ × base × height

To calculate the area of ΔABC, we take any of the sides as base and the corresponding perpendicular from the vertex to the base as the height.

In ΔABC,
AD ⊥ BC, BE ⊥ AC and
CF ⊥ AB.

$ar(\Delta ABC) = \frac{1}{2} \times BC \times AD$

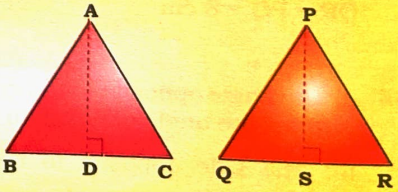
$= \frac{1}{2} \times AC \times BE = \frac{1}{2} \times AB \times CF$



(i) If the height of the two triangles is equal, the ratio of their areas is proportional to the ratio of their bases.

- If AD = PS, then
  Ar (ΔABC) : Ar (ΔPQR) = BC : QR

(ii) If the bases of the two triangles are equal, the ratio of their areas is proportional to the ratio of their height.



- If BC = QR, then
  Ar (ΔABC) : Ar (ΔPQR) = AD : PS

(B) Area of scalene triangle

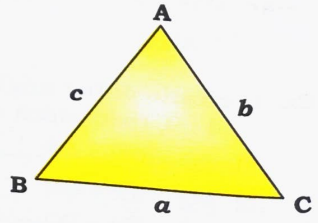
Area of ΔABC

$= \sqrt{s(s-a)(s-b)(s-c)}$

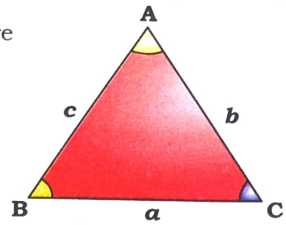
(Heron's formula)

Where, Semi Perimeter (s)

$= \frac{a+b+c}{2}$



(C) When two sides and the angle between these two sides are given



Area of $\Delta ABC = \frac{1}{2}$ ab sinC

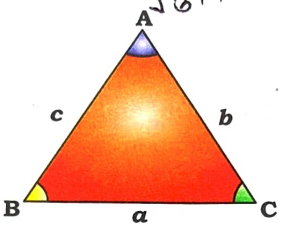
$= \frac{1}{2}$bc sinA $= \frac{1}{2}$ac sinB

Sine & Cosine Rule

(a) Sine Rule

tan 15° = cot 75° = $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

The ratio of side and sine of opposite angle of a triangle is equal to twice of circum-radius.



$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R;$

Where, R is the circumradius of triangle.

sin 15° = cos 75° = $\frac{\sqrt{3}-1}{2\sqrt{2}}$

sin 75° = cos 15° = $\frac{\sqrt{3}+1}{2\sqrt{2}}$

(b) Cosine Rule

If two sides and the angle between those sides are given, then we can find the opposite side by Cosine Rule.

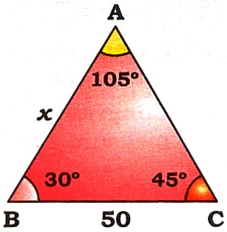
$\cos A = \frac{b^2+c^2-a^2}{2bc}$, $\cos B = \frac{a^2+c^2-b^2}{2ac}$

$\cos C = \frac{a^2+b^2-c^2}{2ab}$

tan 75° = cot 15° = $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

Ex. In ΔABC, $\angle B = 30°$ and $\angle C = 45°$. If BC = 50cm, then find the length of AB.

HINTS



$\Rightarrow \frac{50}{\sin 105°} = \frac{x}{\sin 45°}$

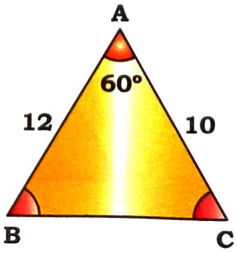
$\Rightarrow \frac{50}{\frac{(\sqrt{3}+1)}{2\sqrt{2}}} = \frac{x}{\frac{1}{\sqrt{2}}}$

$\left(\because \sin 105° = \frac{(\sqrt{3}+1)}{2\sqrt{2}}\right)$

$\Rightarrow x = \frac{100}{(\sqrt{3}+1)} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = 50(\sqrt{3}-1)$

Ex. In ΔABC, AB = 12cm and AC = 10cm, and $\angle BAC = 60°$. What is the length of side BC?

HINTS $\cos A = \frac{AB^2+AC^2-BC^2}{2\times AB\times AC}$



$\Rightarrow \cos 60° = \frac{(12)^2+(10)^2-BC^2}{2\times 12\times 10}$

$\Rightarrow \frac{1}{2} = \frac{144+100-BC^2}{2\times 120}$

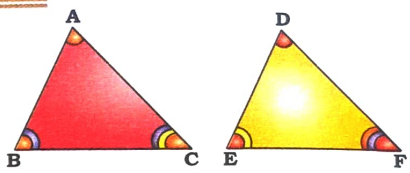
$\Rightarrow 120 = 244 - BC^2$

$\Rightarrow BC^2 = 124 \Rightarrow BC = 2\sqrt{31}$ cm

Congruency & Similarity in Triangles

Congruency in Triangles

- Two triangles are Congruent, if



(i) Their corresponding angles are equal

$\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$

(ii) Their corresponding sides are equal.

AB = DE, BC = EF, CA = FD

Similarity in Triangles

- Two triangles are similar, if

(i) Their corresponding angles are equal

$\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$

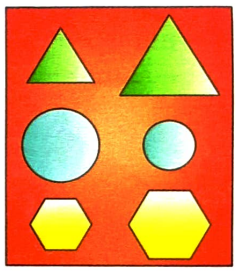
(ii) Their corresponding sides are in the same ratio (or proportion).

$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$



All congruent figures are similar, but not all similar figures are congruent.

Similarity in Regular Figures



Any two figures could be tested to check for similarity and congruence. In the case of regular figures, this is easiest – any two regular figures with the same number of sides will be similar to each other.

For example if we take two regular hexagons, or two circles, or two equilateral triangles, or two squares, or two regular pentagons, each pair of figures will be similar without any further checking required.

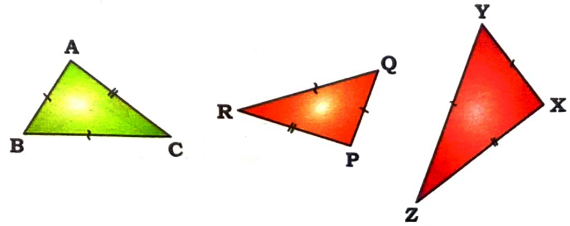
Figures may differ in appearance; test their similarity using following rules.

Conditions for Congruency and Similarity of Triangles

SSS (Side-Side-Side) Test

If we check the three sides of two triangles, then the triangles are

- **Congruent** if three pairs of sides of the two triangles are equal in length.
- **Similar** if the corresponding sides of two triangles have lengths in the same ratio.



In ΔABC and ΔPQR: AB = PQ, BC = QR, AC = PR

$\therefore \Delta ABC \cong \Delta PQR$

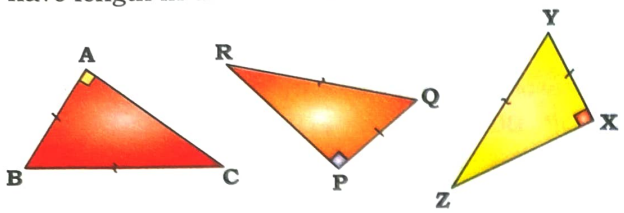
In ΔPQR and ΔXYZ: $\frac{PQ}{XY} = \frac{QR}{YZ} = \frac{PR}{XZ}$

$\therefore \Delta PQR \sim \Delta XYZ$

Hypotenuse Side Test

If we check the sides of two right-angled triangles, then the triangles are

- **Congruent** if the hypotenuses and one of the shorter sides are equal in length.
- **Similar** if the hypotenuses and one pair of shorter sides have length in the same ratio.



In ΔABC and ΔPQR: $AB = PQ$, $BC = RQ$

$\therefore \Delta ABC \cong \Delta PQR$

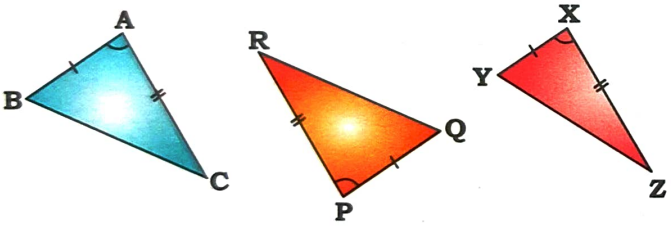
In ΔPQR and ΔXYZ: $\frac{PQ}{XY} = \frac{QR}{YZ}$

$\therefore \Delta PQR \sim \Delta XYZ$

SAS (Side-Angle-Side) Test

If we check two side and its included angle of two triangles then the triangle are

- **Congruent** if the two sides are equal in length and the angle between equal sides is also equal.
- **Similar** if the two sides have lengths in the same ratio and the angle between them is equal.



In ΔABC and ΔPQR: $AB = PQ$, $AC = PR$, $\angle A = \angle P$

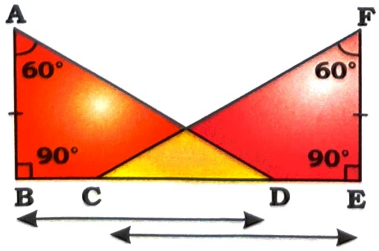
$\therefore \Delta ABC \cong \Delta PQR$

In ΔPQR and ΔXYZ: $\frac{PQ}{XY} = \frac{QR}{YZ}$, $\angle P = \angle X$

$\therefore \Delta PQR \sim \Delta XYZ$

Ex. In ΔABD and ΔFEC, $\angle BAD = 60°$, $l(BD) = l(EC)$, $\angle ABD = \angle FEC = 90°$, and $l(AB) = l(FE)$. Find the ratio of $\angle BAD$ and $\angle FCE$.

HINTS



In ΔABD and ΔFEC,

$AB = FE$ (given)

$BD = CE$ (given)

$\angle B = \angle E = 90°$

$\Delta ABD \cong \Delta FEC$ (from SAS)

$\Rightarrow \angle F = 60°$

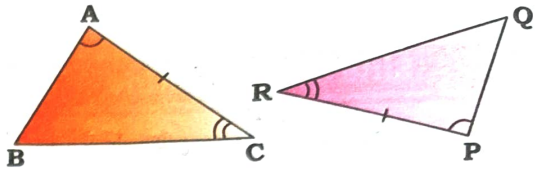
$\therefore \angle FCE = 180° - 90° - 60° = 30°$

$\therefore \frac{\angle BAD}{\angle FCE} = \frac{60°}{30°} = \frac{2}{1} = 2 : 1$

ASA (Angle-Side-Angle) Test

If we check two angles and the included side of two triangles then the triangle are

- **Congruent** if the two pairs of angle have the same measure and only one side are equal in length.

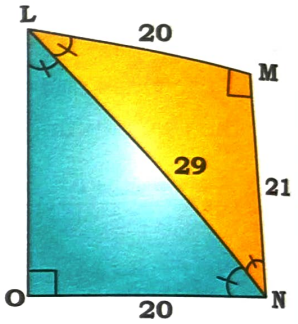


In ΔABC and ΔPQR: $BC = QR$, $\angle A = \angle P$, $\angle C = \angle R$

$\Delta ABC \cong \Delta PQR$

Ex. ΔLON and ΔLMN are two right-angled triangles with common hypotenuse LN such that $\angle LON = 90°$ and $\angle LMN = 90°$. LN is the bisector of $\angle OLM$. If $LN = 29$ cm and $ON = 20$ cm, then what is the perimeter (in cm) of ΔLMN?

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In ΔLON and ΔLMN,

$\angle LON = \angle LMN = 90°$

$LN = LN$ (common side)

$\angle OLN = \angle MLN$ (angle bisector)

$\therefore \Delta LON \cong \Delta LMN$ (By ASA)

In ΔLMN,

$MN^2 = 29^2 - 20^2 \Rightarrow MN = 21$

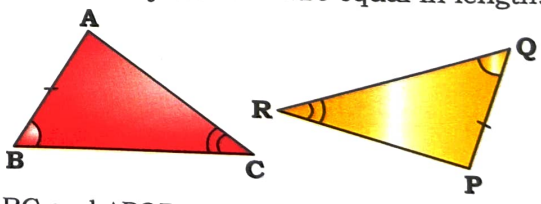
$\therefore$ Perimeter of $\Delta LMN = 29 + 20 + 21$

$= 70$ cm

AAS (Angle-Angle-Side) Test

If we check two angles and a corresponding non included side of two triangles then the triangle are

- **Congruent** if the two pairs of angles have the same measure and only one side are equal in length.



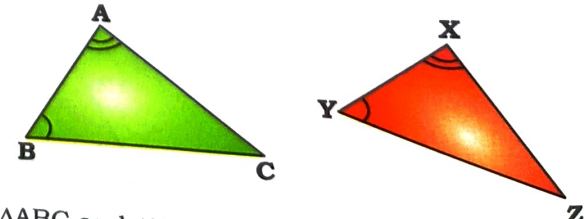
In ΔABC and ΔPQR: $AB = PQ$, $\angle B = \angle Q$, $\angle C = \angle R$

$\Delta ABC \cong \Delta PQR$

AA Test

If we check the angles of two triangles then the triangles are

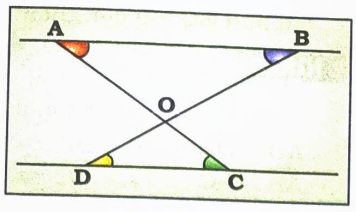
- Similar if two angles of triangle are equal then both triangle are similar.



In ΔABC and ΔPQR: $\angle A = \angle X$, $\angle B = \angle Y$

$\Delta ABC \sim \Delta XYZ$

Spotting Similarity & Congruency



- Identifying similarity (and congruency) is crucial in geometry, especially for visualizing problems.
- A key to recognizing similarity is spotting equal angles.

Case 01: Parallel Lines

- When you see two parallel lines intersected by transversals, immediately look for similar triangles.
- This is because parallel lines create many pairs of equal angles (e.g., alternate interior angles, corresponding angles).

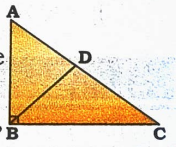
Given parallel lines AB and CD, and transversals AC and BD intersecting at O:

- Triangles ΔAOB and ΔCOD are similar because they have two pairs of equal angles:
  - ➤ ∠OAB = ∠OCD (Alternate interior angles)
  - ➤ ∠OBA = ∠ODC (Alternate interior angles)
  - ➤ ∠AOB = ∠COD (Vertically opposite angles)

Case 02: Altitude to the Hypotenuse in a Right Triangle:

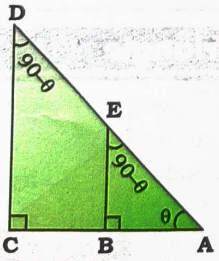
When you draw an altitude from the right angle vertex to the hypotenuse of a right-angled triangle, it creates three similar triangles:

- The original right triangle.
- The two smaller triangles formed by the altitude.
- Example (using ΔABC right-angled at B, with BD as altitude to AC):

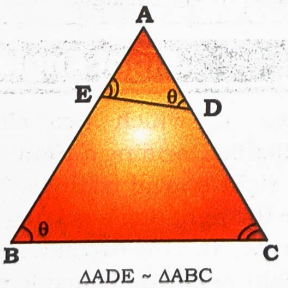


- ΔABC ~ ΔADB (They share ∠A, and both have a right angle).
- ΔABC ~ ΔBDC (They share ∠C, and both have a right angle).
- Therefore, all three triangles are similar to each other: ΔABC ~ ΔADB ~ ΔBDC.

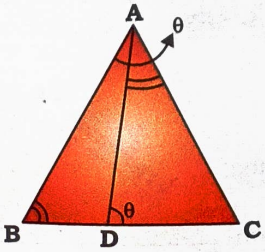
Some Similar Figure



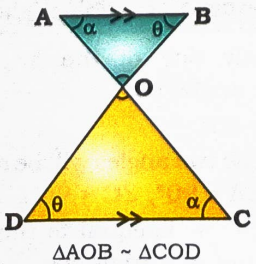
ΔABE ~ ΔACD



ΔADE ~ ΔABC



∠ADC = ∠BAC = θ
∠DAC = ∠ABC (Given)
ΔABC ~ ΔADC



ΔAOB ~ ΔCOD

Ex. In ΔABC, ∠B = 87° and ∠C = 60°. Points D and E are on the sides AB and AC, respectively, such that ∠DEC = 93° and DAE is

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∠DEA = ∠ABC

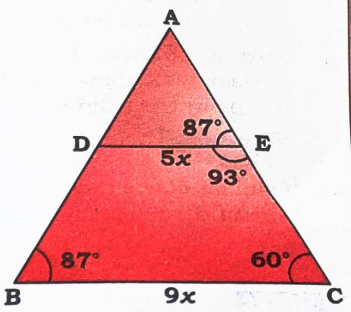
∠A = Common

⇒ ΔAED ~ ΔABC

$\Rightarrow \frac{AE}{AB} = \frac{ED}{BC} = \frac{AD}{AC}$

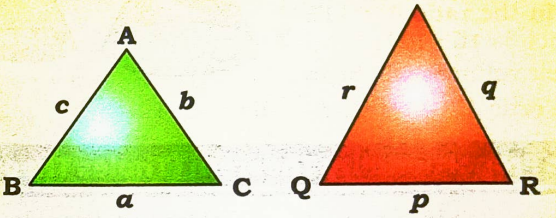
$\Rightarrow \frac{AE}{14.4} = \frac{5}{9}$

⇒ AE = 8 cm



Properties in Similar triangles

If ΔABC and ΔPQR are similar, then



(i) $\frac{a}{p} = \frac{b}{q} = \frac{c}{r}$

(ii) Ratio of corresponding sides
= Ratio of perimeter
= Ratio of semi-perimeter(s)
= Ratio of corresponding medians
= Ratio of inradius
= Ratio of circumradius

(iii) Ratio of area
= Ratio of square of corresponding sides
= Ratio of square of perimeter
= Ratio of square of semi-perimeter
= Ratio of square of corresponding medians
= Ratio of square of inradius
= Ratio of square of circumradius

Ex. If the ratio of corresponding sides of two similar triangles is $\sqrt{3} : \sqrt{2}$, then what is the ratio of the area of the two triangles?

HINTS $ar(\Delta ABC):ar(\Delta PQR) = (\sqrt{3})^2 : (\sqrt{2})^2 = 3 : 2$

Ex. Let ΔABC ~ ΔQPR and $\frac{ar(\Delta ABC)}{ar(\Delta QPR)} = \frac{64}{169}$. If AB = 10 cm, BC = 7 cm and AC = 16 cm, then QR (in cm) is equal to:

HINTS Here, ΔABC ~ ΔQPR & $\frac{ar(\Delta ABC)}{ar(\Delta QPR)} = \frac{64}{169}$

$\Rightarrow \frac{AC}{QR} = \sqrt{\frac{64}{169}} \Rightarrow \frac{16}{QR} = \frac{8}{13}$

$\Rightarrow QR = \frac{13 \times 16}{8} = 26$ cm

Similarity of Cones

If two cones are similar and their heights (a linear dimension) are in the ratio $x : y$.

- **Linear Dimensions:** Their other linear dimensions (like base radii, slant heights) will also be in the same ratio $x : y$.
- **Surface Areas:** Their various surface areas (total surface area, base surface area, curved surface area) will be in the ratio $x^2 : y^2$.
- **Volumes:** Their volumes will be in the ratio $x^3 : y^3$.

Ex. If similar cones have heights in the ratio 4 : 11. Find the ratio of the following:

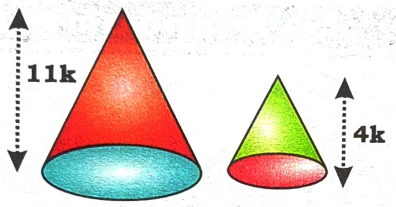
Slant height, surface area and volume

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Their base radii and slant heights will also be in the ratio 4 : 11.

Their surface areas will be in the ratio $4^2 : 11^2$, which is 16 : 121.

Their volumes will be in the ratio $4^3 : 11^3$, which is 64 : 1331.



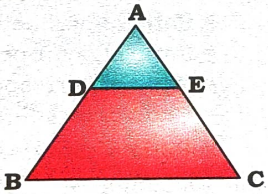
Basic Proportionality Theorem (Thales Theorem)

- A line drawn parallel to one side of a triangle divides other two sides in the same ratio.
- If a line divides any two sides of a triangle in the same ratio, the line must be parallel to the third side.

In ΔABC, if DE || BC, then

$$\frac{AD}{DB} = \frac{AE}{EC}$$

'OR' If $\frac{AD}{DB} = \frac{AE}{EC}$, then DE || BC



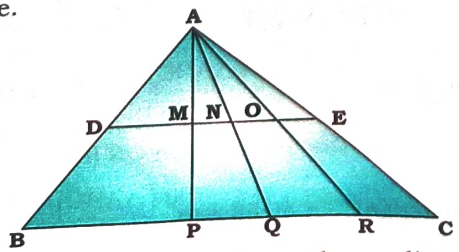
- Some of the results derived from this theorem, we will use, are as follows:

(i) $\frac{AD}{BD} = \frac{AE}{EC}$ (ii) $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC}$

(iii) $\Delta ADE \sim \Delta ABC$

(iv) $\frac{Ar(\Delta ADE)}{Ar(\Delta ABC)} = \left(\frac{AD}{AB}\right)^2 = \left(\frac{AE}{AC}\right)^2 = \left(\frac{DE}{BC}\right)^2$

- A line drawn parallel to one side of a triangle divides the median, the angle bisector and the altitude of triangle in the same ratio it divides the other two sides of the triangle.



In ΔABC, AP, AQ and AR are the median, the angle bi-sector and altitude respectively and DE||BC, then

$$\frac{AD}{DB} = \frac{AE}{EC} = \frac{AM}{MP} = \frac{AN}{NQ} = \frac{AO}{OR}$$

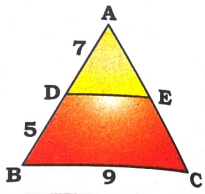
Ex. In ΔABC, D and E are the points on sides AB and AC, respectively such that $\angle ADE = \angle B$. If AD = 7 cm BD = 5 cm and BC = 9 cm, then DE (in cm) is equal to____

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Using Basic proportionality theorem,

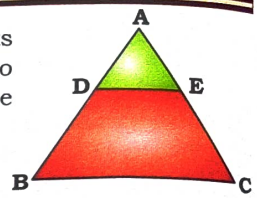
$$\frac{AD}{AB} = \frac{DE}{BC} \Rightarrow \frac{7}{12} = \frac{DE}{9}$$

$$\Rightarrow DE = \frac{63}{12} = 5.25 \text{ cm}$$



Mid-Point Theorem

The line segment joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the magnitude of third side and vice-versa.



- If D and E are mid-points of AB and AC, respectively, then DE || BC and $DE = \frac{BC}{2}$
- DE || BC and $DE = \frac{BC}{2}$, then D and E are the mid-points of AB and AC respectively. In this case

(i) $\frac{AD}{AB} = \frac{AE}{AC} = \frac{DE}{BC} = \frac{1}{2}$ (ii) $\frac{AD}{DB} = \frac{AE}{EC} = 1$

(iii) $\Delta ADE \sim \Delta ABC$ (iv) $\frac{Ar(\Delta ADE)}{Ar(\Delta ABC)} = \frac{1}{4}$

Types of Triangle

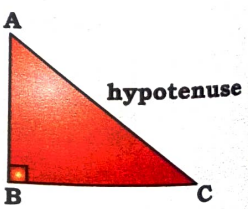
Triangles are classified on the basis of angles and sides–

Types of Triangle

On the basis of Angle	On the basis of Sides
• Right angle triangle	• Scalene triangle
• Acute angle triangle	• Isosceles triangle
• Obtuse angle triangle	• Equilateral triangle

Right Angle Triangle

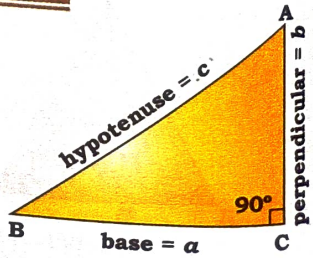
A triangle in which two sides are perpendicular, forming a right angle, is called a right-angled triangle. The side opposite the right angle is known as the hypotenuse, and the two sides adjacent to the right angle are called legs.



Properties of a Right angle Triangle

(i) Exactly one of the angle is right angle, i.e.
$\angle B = 90°$

- Exactly two angles will be acute. i.e. $\angle A < 90°$, $\angle C < 90°$
- One angle is equal to the sum of other two angle, i.e.
$\angle B = \angle A + \angle C = 90°$



(ii) (a) **Pythagoras Theorem:** In a right triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

If a, b and c be three sides of a right-angled triangle, then according to the Pythagoras Theorem,

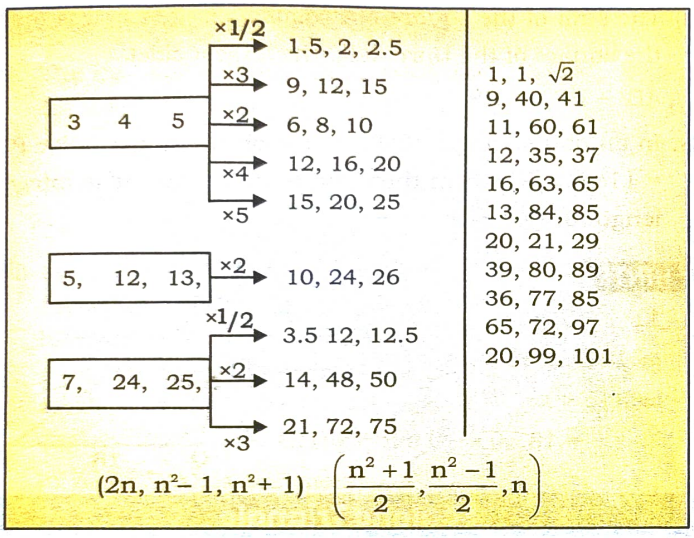
$$a^2 + b^2 = c^2$$

For Example: $3^2 + 4^2 = 5^2$

(b) **Pythagorean Triplets:** A set of three integers a, b, c which satisfy Pythagoras Theorem ($a^2 + b^2 = c^2$) or are the sides of a right angle triangle is called Pythagorean triplets.

- The smallest Pythagorean triplet is (3, 4, 5).
- If (a, b, c) be a Pythagorean triplets, then (ak, bk, ck) or $\left(\frac{a}{k}, \frac{b}{k}, \frac{c}{k}\right)$ will also be the Pythagorean triplet.

Following Pythagorean triplets are frequently used in the examinations.



Tricks for Making Triplets

(a) For even number →

$6 \to \frac{6}{2} = 3 \xrightarrow{\text{Square}} 9 \begin{cases} \xrightarrow{-1} 8 \\ \xrightarrow{+1} 10 \end{cases}$

$8 \to \frac{8}{2} = 4 \xrightarrow{\text{Square}} 16 \begin{cases} \xrightarrow{-1} 15 \\ \xrightarrow{+1} 17 \end{cases}$

$12 \to \frac{12}{2} = 6 \xrightarrow{\text{Square}} 36 \begin{cases} \xrightarrow{-1} 35 \\ \xrightarrow{+1} 37 \end{cases}$

(b) For odd number →

$3 \xrightarrow{\text{Square}} 9 \to \frac{9}{2} = 4.5 \begin{cases} \xrightarrow{-0.5} 4 \\ \xrightarrow{+0.5} 5 \end{cases}$

$5 \xrightarrow{\text{Square}} 25 \to \frac{25}{2} = 12.5 \begin{cases} \xrightarrow{-0.5} 12 \\ \xrightarrow{+0.5} 13 \end{cases}$

$7 \xrightarrow{\text{Square}} 49 \to \frac{49}{2} = 24.5 \begin{cases} \xrightarrow{-0.5} 24 \\ \xrightarrow{+0.5} 25 \end{cases}$

"(20, 21, 29) and (20, 99, 101) are triplets."

Ex. ΔABC is right angled at B and D is a point on BC such that BD = 5 cm, AD = 13 cm and AC = 37 cm, then find the length of BC in cm.

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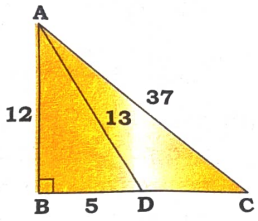
AB = 12 cm (By triplet of 5, 12, 13)

$(12)^2 + (BC)^2 = (37)^2$

$\Rightarrow (BC)^2 = (37)^2 - (12)^2$

$\Rightarrow BC^2 = 25 \times 49$

$\Rightarrow BC = 5 \times 7 = 35$



Alternatively

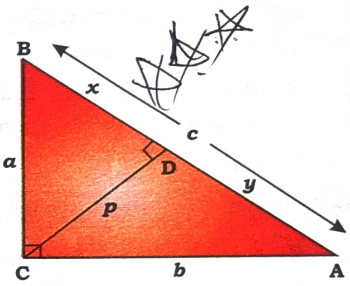
(5, 12, 13) and (12, 35, 37) are triplet.

With the help of triplet, AB = 12cm satisfy the given condition.

∴ BC = 35 cm

(iii) In a right angle ΔABC, if a perpendicular (CD) is drawn from the vertex which is right angle (∠C) to the hypotenuse (AB), then,

ΔABC ~ ΔBCD ~ ΔACD



- $a^2 = x \times c$
- $p^2 = x \times y$
- $b^2 = y \times c$
- $a \times b = p \times c$
- $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

Ex. In ΔABC, ∠A = 90°, AD⊥BC at D. If AB = 12 cm and AC = 16 cm, then what is the length (in cm) of BD?

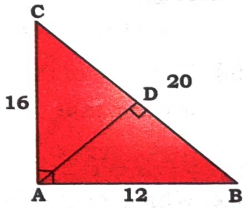
HINTS

$AB^2 = BD \times BC$

$\Rightarrow 12^2 = BD \times 20$

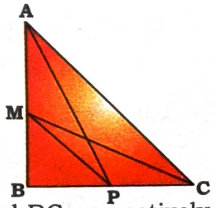
$\Rightarrow 144 = BD \times 20$

$\Rightarrow BD = 7.2cm$



(iv) If P and M are the two points on the sides BC and AB respectively of ΔABC, right-angled at B, then

$AP^2 + MC^2 = AC^2 + MP^2$



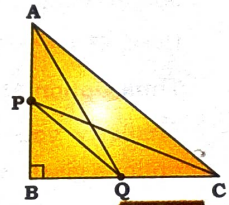
Ex. Points P and Q are on the sides AB and BC respectively of a ΔABC, right angled at B. If AQ = 11 cm, PC = 8 cm, and AC = 13 cm, then find the length (in cm) of PQ.

HINTS

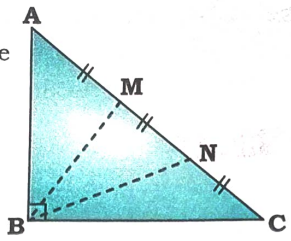
$AQ^2 + PC^2 = AC^2 + PQ^2$

$\Rightarrow 121 + 64 = 169 + PQ^2$

$\Rightarrow 16 = PQ^2 \Rightarrow PQ = 4$



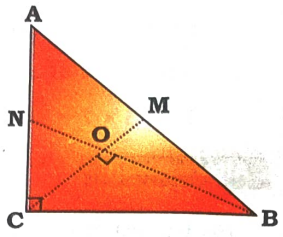
(v) ΔABC is a right angle triangle, right angle at B. If M, N are points on AC such that AM = MN = NC then



$$BM^2 + BN^2 = \frac{(n-1)(2n-1)AC^2}{6n}$$

Where, n = No. of equal parts

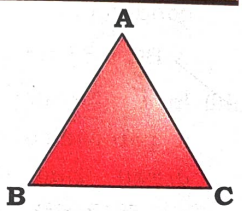
(vi) In ΔABC, $\angle C = 90°$, M and N are mid-points of sides AB and AC respectively. CM and BN intersects each other at O and $\angle BOC$ is 90° then $BN = \frac{BC\sqrt{6}}{2}$



Acute Angle Triangle

If each of the angles of a triangle is acute, then the triangle is called an acute-angled triangle.

If $\angle A < 90°$, $\angle B < 90°$ and $\angle C < 90°$, then ΔABC is an acute-angled triangle.



Properties of an acute angle triangle

(i) The sum of any two angles of the triangle is always greater than the third angle.

$\angle A + \angle B > \angle C$, $\angle B + \angle C > \angle A$, $\angle C + \angle A > \angle B$

Ex. Which one of the following cannot be the ratio of angles in acute angle triangle?

(a) 2 : 5 : 8 (b) 4 : 1 : 4

(c) 2 : 3 : 4 (d) 1 : 1 : 1

HINTS Using property (i), Option (a) can't be the ratio of angles of an acute angle triangle.

(ii) The sum of the squares of any two sides of the triangle is greater than the square of the third side.

$AB^2 + BC^2 > AC^2$, $BC^2 + AC^2 > AB^2$, $AC^2 + AB^2 > BC^2$

Ex. Which of the following can be the 3 sides of an acute angle triangle?

(a) 4 cm, 8 cm, 7 cm

(b) 1 cm, 2 cm, 3 cm

(c) 2 cm, 4 cm, 3 cm

(d) 9 cm, 12 cm, 15 cm

HINTS In such type of question you have to check each option using property (ii).

Here, $4^2 + 8^2 > 7^2$, $8^2 + 7^2 > 4^2$ & $7^2 + 4^2 > 8^2$

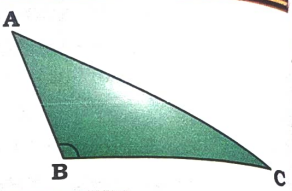
Thus Option(A) is right answer.

Similarly you can check other options but all other options given will not satisfy the property.

Obtuse Angle Triangle

If one of the angle of a triangle is obtuse, then the triangle is called an obtuse-angled triangle.

ΔABC is obtuse angle triangle.



Properties of an obtuse-angled triangle

(i) Exactly one of the angle of the triangle is obtuse and other two angles are acute.

$\angle B > 90°$ and $0° < \angle A, \angle C < 90°$

(ii) The sum of the two acute angles of the triangle is less than the obtuse angle.

$(\angle A + \angle C) < \angle B$

(iii) The sum of the square of two smaller sides is less than the square of the third side (the largest side).

- $AB^2 + BC^2 < AC^2$

Ex. In an obtuse angle ΔPQR, $\angle Q$ is obtuse angle if side PQ = 11cm, QR =15cm then find minimum possible integer length of side PR?

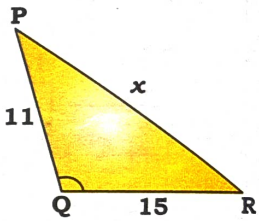
HINTS

$11^2 + 15^2 < x^2$

$\Rightarrow 121 + 225 < x^2$

$\Rightarrow 346 < x^2$

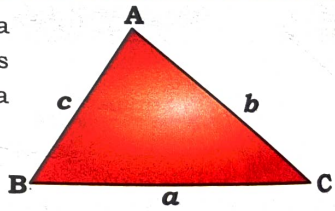
$\Rightarrow x_{min} = 18.60 \sim 19$ cm



Scalene Triangle

If all of the three sides of a triangle are of different lengths then the triangles is called a scalene triangle.

$AB \neq BC \neq CA \Rightarrow a \neq b \neq c$



Properties of a scalene triangle

(i) No two sides are equal in length.

$AB \neq BC \neq CA$ or $a \neq b \neq c$

(ii) No two angles are equal, i.e., $\angle A \neq \angle B \neq \angle C$

(iii) Triangle may be one of the acute angled, right angled or obtuse angled.

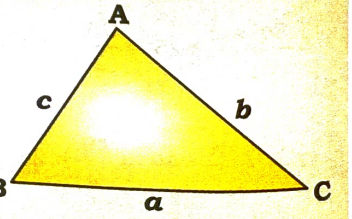
(iv) **Perimeter, Semi-perimeter and Area of scelene triangle**

- Perimeter = a + b + c
- Semi-perimeter(s)

$$= \frac{a+b+c}{2}$$

- Area of ΔABC

$$= \sqrt{s(s-a)(s-b)(s-c)}$$



Ex. The sides of the scalene triangle ABC are in the ratio 3 : 5 : 6 and the semi-perimeter is 42 cm, what is the difference of the largest and the smallest sides of the triangle?

HINTS Semi-perimeter = $\frac{3x+5x+6x}{2}$

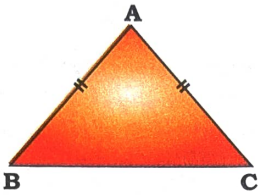
$\Rightarrow 42 = 7x \Rightarrow x = 6$

Required Difference = $6x - 3x = 3 \times 6 = 18$ cm

Isosceles Triangle

If two sides of a triangle are equal in length, then the triangle is called an isosceles triangle.

If AB = AC ≠ BC, then ΔABC is an isosceles triangle.



Properties of an isosceles triangle

(i) The length of the two sides are equal, but the length of third side can be smaller or larger than the equal sides.

AB = AC & BC > AB or BC < AB

(ii) Two angles opposite to the equal sides are equal.

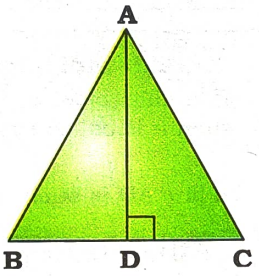
∠B = ∠C (opposite to the side AC and AB)

(iii) In an isosceles ΔABC, the median, the angle bisector, perpendicular bisector and the altitude of unequal side is the same and it divides ΔABC in two congruent (equal) right-angled triangle.

In ΔABC, AD is the median, the angle bisector, perpendicular bisector and the altitude.

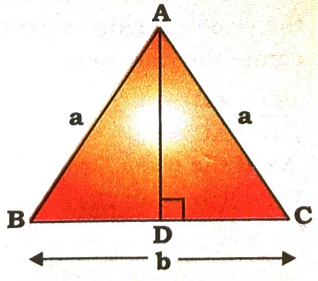
AD ⊥ BC and BD = DC, then

ΔABD ≅ ΔACD



(iv) **Perimeter, semi-perimeter and Area**

- Perimeter = AB + BC + CA = 2a + b.
- Semi perimeter $= \frac{AB+BC+CA}{2} = a + \frac{b}{2}$
- AD = $\frac{\sqrt{4a^2-b^2}}{2}$
- Area of ΔABC = $\frac{b}{4}\sqrt{4a^2-b^2}$



In isosceles triangle ABC, ΔABD and ΔACD are the two congruent right-angled triangle. Use Pythagorean triplets in these right angled triangle.

Ex. The perimeter of an isosceles triangle is 50 cm. If the base is 18 cm, then find the length of equal sides.

HINTS Perimeter, 2a + b = 50

$\Rightarrow$ 2a + 18 = 50

$\Rightarrow$ a = 16

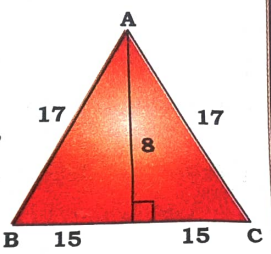
Ex. The length of perpendicular drawn on the base of an isosceles triangle is 8 cm. If its perimeter is 64 cm, what is the area of this triangle?

HINTS

Check 30 + 34 = 64

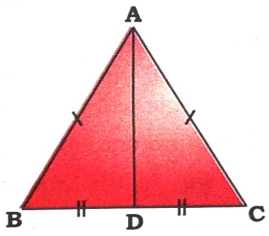
[Apply concept of triplet, (8, 15, 17) to solve quickly]

Area = $\frac{1}{2} \times 30 \times 8 = 120$ cm²



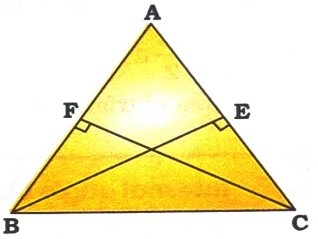
(v) If a line (which joins the common vertex of two equal sides of a triangle) bisects the base then that line is perpendicular to the base and vice versa.

If AB = AC and BD = DC, then ∠ADB = ∠ADC = 90°

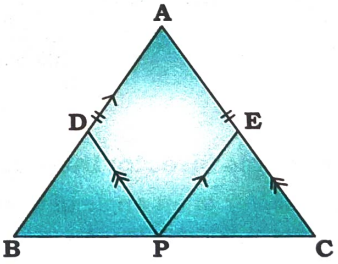


(vi) • If AB = AC and BE ⊥ AC and CF ⊥ AB, then BE = CF

• If AB = AC, E and F are the mid-points, then BE = CF

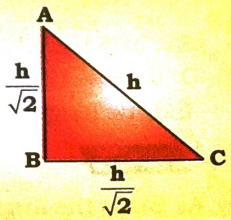


(vii) ΔABC is an isosceles triangle (AB = AC). If P is a point on the side BC and DP || AC and EP || AB then DP + PE = AB = AC



Isosceles Right Angle Triangle

- Ratio of sides
  $\Rightarrow$ AB : BC : AC = 1 : 1 : $\sqrt{2}$
- Area = $\frac{(\text{Hypotenuse})^2}{4}$
- Perimeter = $h(\sqrt{2}+1)$



Ex. If the hypotenuse of the isosceles right angle triangle equals the side of a square whose area is 256 cm². Find the area of triangle.

HINTS Side of square = $\sqrt{256} = 16$.

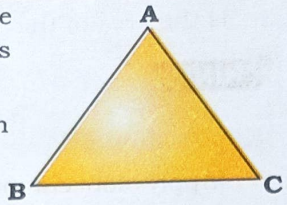
$\therefore$ side of square = $h = 16$

Area = $\frac{h^2}{4} = \frac{16^2}{4} = \frac{256}{4} = 64$ cm².

Equilateral Triangle

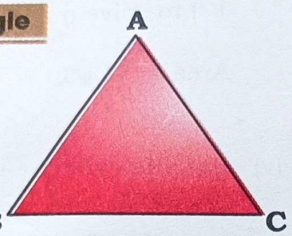
If all the three sides of a triangle are equal in length, then the triangle is called an equilateral triangle.

If AB = BC = AC, then ΔABC is an equilateral triangle.



Properties of an equilateral triangle

Let ΔABC is an equilateral triangle in which AB = BC = AC



(i) The length of all the three sides are equal, i.e. AB = BC = AC

(ii) All the three angles will be equal and the value of each angle will be 60° i.e. ∠A = ∠B = ∠C

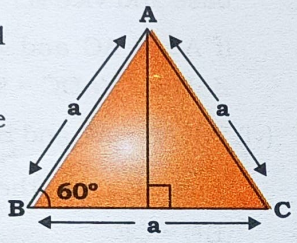
(iii) In an equilateral ΔABC,

(a) All the three medians are equal

(b) All the three altitudes are equal

(c) Each median = Each altitude $= \frac{\sqrt{3}}{2}a$

(d) Area of triangle = $\frac{\sqrt{3}}{4}a^2$



(iv) In equilateral ΔABC, incentre, circumcentre, orthocentre and centroid are at the same point.

(v) If altitudes of a triangle are equal, the triangle is an equilateral triangle.

Centres of Triangle

We will study 5 types of centres of a triangle

(i) Incentre (ii) Circumcenter

(iii) Orthocentre (iv) Centroid

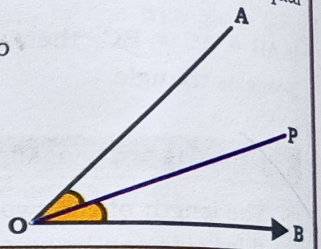
(v) Excentre

Angle Bisector: A line which divides an angle in two equal parts is called the angle bisector.

If line OP divides ∠AOB in two equal parts then

$\angle AOP = \angle BOP = \frac{\angle AOB}{2}$, then

OP is an angle bisector.



Angle bisector of an angle of a triangle

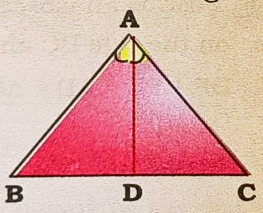
- Interior or Internal
- Exterior or External

Internal angle bisector of a triangle

If a line bisects the Internal angle of a triangle, then it is called internal angle bisector of a triangle.

If line AD is such that

∠BAD = ∠CAD, then AD is the interior bisector of ∠A.



Internal angle bisector theorem

The Internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the sides containing the angle, i.e.

$\frac{BD}{CD} = \frac{AB}{AC}$

Ex. In ΔABC, the bisector of ∠B meets AC at point D. If AB = 12 cm, BC = 18 cm and AC = 15 cm, then find the length (in cm) of AD.

HINTS

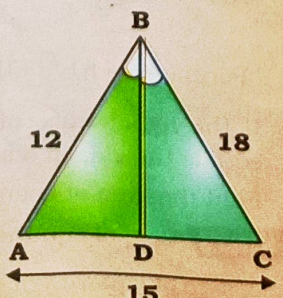
As BD is the angle bisector of ∠ABC

$\Rightarrow \frac{AB}{BC} = \frac{AD}{DC} \Rightarrow \frac{12}{18} = \frac{AD}{DC}$

$\Rightarrow$ AD : DC = 2 : 3

Also, AC = 15 cm

$\therefore$ AD = $\frac{15}{(2+3)} \times 2 = 6$ cm



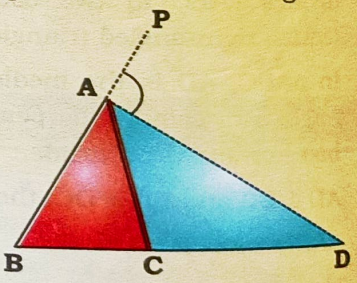
External angle bisector of a triangle

If a line bisects the exterior angle of a triangle, then it is called external angle bisector of a triangle.

If line AD is such that

$\angle PAD = \angle DAC = \left(\frac{\pi - \angle A}{2}\right)$,

then AD is the external bisector of ∠A.

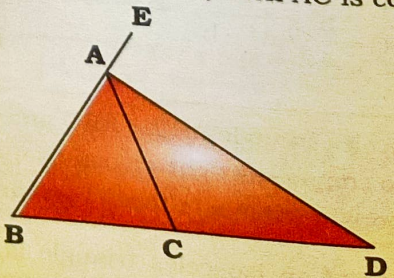


External angle bisector theorem

The external bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle

$\frac{BD}{CD} = \frac{AB}{AC}$

Ex. If the given figure, 2∠EAD = ∠EAC, BC = 40 cm, BA = 8 cm and CD = 24 cm, then AC is equal to



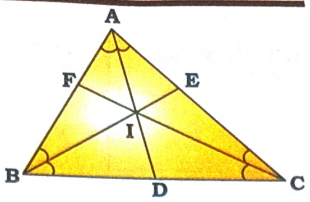
HINTS

Since AD is angle bisector

$\frac{AB}{AC} = \frac{BD}{CD} \Rightarrow \frac{8}{AC} = \frac{64}{24} \Rightarrow AC = 3$ cm

Incentre

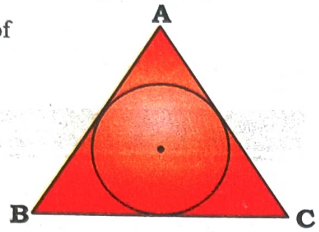
The point of intersection of the internal bisector of the angles of a triangle is called the incentre.



- If AD, BE and CF are the internal angle bisectors of the $\angle A$, $\angle B$ and $\angle C$ respectively, then all the three bisectors of the angles pass through a point, which is called the incentre of the ΔABC.
- The incentre is the centre of the incircle of a triangle.
- It is generlly denoted by 'I'.

  I → Incentre of ΔABC

  I → Centre of the circle



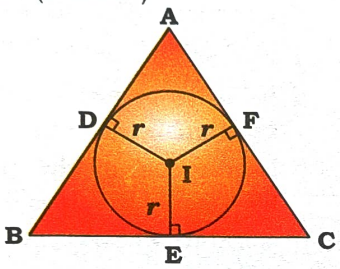
Properties of incentre

(i) All the three internal bisectors of the angles of a triangle pass through a common point (incentre).

(ii) The incentre of the triangle is equidistant from all the three sides of the triangle and the distance is equal to the inradius.

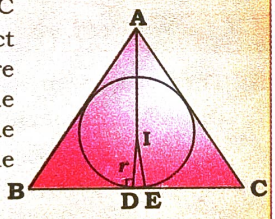
If circle touches the side AB, BC and AC at points D, E and F respectively and I be the incentre, then

ID = IE = IF (inradius)





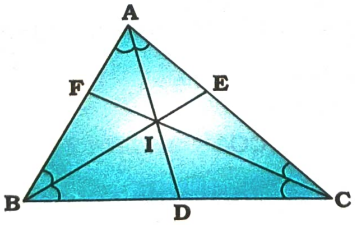
The bisector of $\angle A$ of ΔABC may or may not intersect side BC at point E where the incircle touches the side BC of the triangle and the same is true for other angle bisectors.



The bisector of $\angle A$ meets BC at D and the side BC touches the circle at E.

(iii)

- $\angle BIC = 90° + \frac{\angle A}{2}$
- $\angle AIC = 90° + \frac{\angle B}{2}$
- $\angle AIB = 90° + \frac{\angle C}{2}$



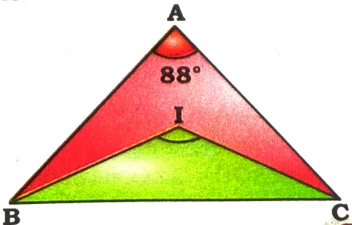
Ex. In ΔABC, $\angle A = 88°$. If I is the incentre of the triangle, then the measure of $\angle BIC$ is:

HINTS

$$\angle BIC = 90° + \frac{\angle A}{2}$$

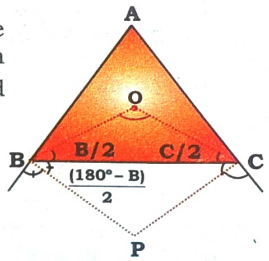
$$= 90° + \frac{88°}{2}$$

$$= 90° + 44° = 134°$$



(iv) The angle between the external bisectors of two angles of a triangle is equal to the difference between right angle and half of the third angle.

$$\angle BPC = 90° - \frac{\angle A}{2}$$





If in ΔABC, the internal bisectors of $\angle B$ and $\angle C$ meet at O and the external bisectors of $\angle B$ and $\angle C$ meet at P, then $\angle BOC + \angle BPC = 180°$.

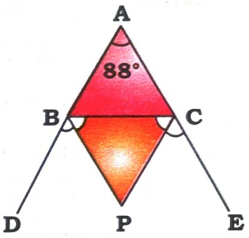
Ex. The sides of ΔABC, AB and AC are extended to the points D and E, respectively. The bisectors of $\angle CBD$ and $\angle BCE$ meet at P. If $\angle A = 88°$, then what will be the measure of $\angle P$?

HINTS

$$\angle BPC = 90° - \frac{1}{2}\angle A$$

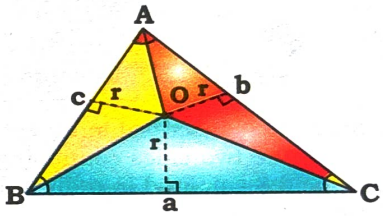
$$\angle BPC = 90° - \frac{1}{2} \times 88°$$

$$\angle BPC = 46°$$

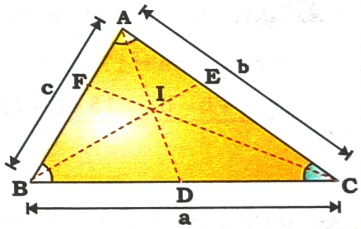


(v) The ratio of area of triangle formed by incentre and two vertex are in ratio in their corresponding sides.

Ar ΔBOC : Ar ΔAOC : Ar ΔAOB = a : b : c



(vi) Incentre divides each angle bisector in the ratio of sum of length of two adjacent sides and opposite side.



Here, AD, BE and CF are angle bisectors of $\angle A$, $\angle B$ and $\angle C$ respectively.

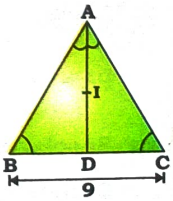
AI : ID = b + c : a BI : IE = a + c : b

CI : IF = a + b : c

Ex. The perimeter of ΔABC is 24 cm and its side, BC = 9 cm. AD is the bisector of $\angle BAC$, while I is the incentre AI : ID is equal to:

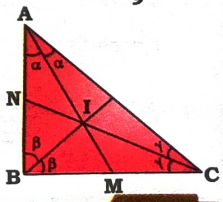
HINTS

$$\frac{AI}{ID} = \frac{AB + AC}{BC} = \frac{15}{9}$$



(vii) ABC is a right angle triangle at B. If BN and BM are given and I is the incentre of triangle then

$BI^2 = BN \times BM$.



In-radius

In-radius of any triangle	In-radius of right angle triangle	In-radius of equilateral triangle
$r = \frac{\Delta}{s} = \frac{\text{Area}}{\text{Semi perimeter}}$	$r = \frac{p+b-h}{2}$	$r = \frac{a}{2\sqrt{3}} = \frac{\text{Side}}{2\sqrt{3}}$

Ex. In ΔABC, perimeter is 24 cm and in-radius is 7 cm. Find the area of ΔABC.

HINTS $s = 24$ cm, $r = 7$cm

$$r = \frac{ar\Delta ABC}{\text{Semi perimeter}} \Rightarrow 7 = \frac{ar\Delta ABC}{12}$$

$\Rightarrow$ ar $\Delta ABC = 84$ cm²

Ex. Find in-radius of a triangle whose sides are 13 cm, 14cm and 15 cm.

HINTS $s = \frac{13+14+15}{2} = 21$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{21\times8\times7\times6}$$

$$= \sqrt{3\times7\times4\times2\times7\times2\times3}$$

$= 3 \times 7 \times 2 \times 2 = 84$ cm²

Now, $r = \frac{\Delta}{s} = \frac{84}{21} = 4$ cm



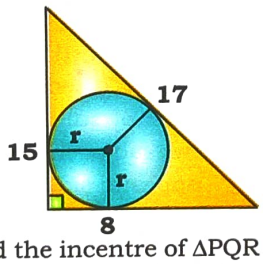
Sides = 13,14,15
Area = 84
Inradius = 4

Ex. A circle inscribed in a right-angled triangle. The lengths of the two sides containing the right angles are 15 cm and 8 cm. Find in radius?

HINTS By triplet of 8, 15 , 17

Hypotenuse = 17 cm

$$r = \frac{p+b-h}{2} = \frac{15+8-17}{2} = 3 \text{ cm}$$



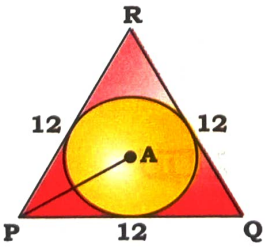
Ex. ΔPQR is an equilateral triangle and the incentre of ΔPQR is point A. If the side of the triangle is 12 cm, then what is the length of PA?

HINTS

$$r = \frac{a}{2\sqrt{3}} = \frac{12\times\sqrt{3}}{2\sqrt{3}\times\sqrt{3}}$$

$= 2\sqrt{3}$ cm

PA $= 2 \times r = 2 \times 2\sqrt{3} = 4\sqrt{3}$ cm

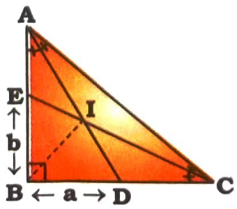


☞ **Special cases:**

Case-I: In right angle triangle ABC,

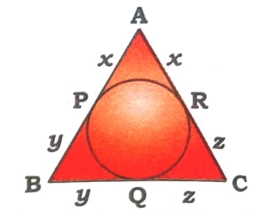
r = radius of ΔABC

$$r = \frac{BI}{\sqrt{2}} = \frac{\sqrt{ab}}{\sqrt{2}}$$



Case-II:

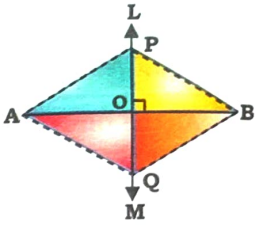
- Semi-perimeter (s) = $x + y + z$
- Area of $\Delta ABC = \sqrt{(x+y+z)xyz}$
- Inradius (r) = $\sqrt{\frac{xyz}{(x+y+z)}}$



Circumcentre

Perpendicular bisector: If a line bisects a line segment perpendicularly, the line is called a perpendicular bisector.

If the line LM ⊥ AB and AO = OB, then AP = PB and AQ = BQ



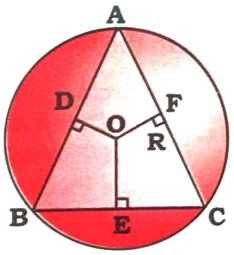
LM is the perpendicular bisector of AB

Circumcentre: The point of intersection of the perpendicular bisectors of the sides of a triangle is called the circumcentre.

$OD \perp AB$ and $AD = BD$

$OE \perp BC$ and $BE = EC$

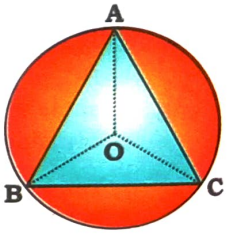
$OF \perp AC$ and $CF = FA$



Properties of Circumcentre

(i) The circumcentre of a triangle is equidistant from all the three vertices of the triangle and the distance is equal to the circumradius, i.e.

OA = OB = OC = R (circumradius)



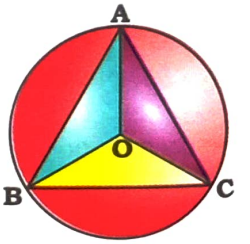
(ii) If O be the circumcentre, then

- In acute-angled triangle ABC,

$\angle BOC = 2\angle A$,

$\angle COA = 2\angle B$ &

$\angle AOB = 2\angle C$

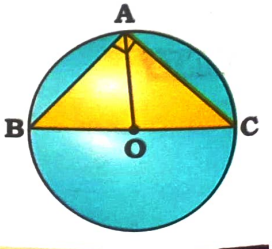


- In a right-angled triangle ABC,

$\angle BOC = 2\angle A = 180°$,

$\angle COA = 2\angle B = 180° - 2\angle C$,

$\angle AOB = 2\angle C = 180° - 2\angle B$

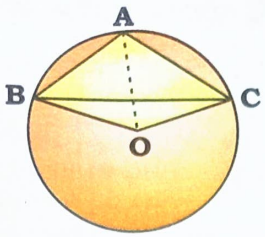


- In an obtuse-angled triangle ABC,

$\angle BOC = 2(180° - \angle A)$,

$\angle COA = 2\angle B$ &

$\angle BOA = 2\angle C$



Ex. From the circumcentre L of ΔXYZ, perpendicular LM is drawn on side YZ. If $\angle YXZ = 60°$, then the measure of $\angle YLM$ is:

HINTS $\angle YXZ = 60°$

LM ⊥ YZ (Given)

$\Rightarrow \angle YLZ = 2 \times \angle YXZ = 120°$

In ΔYLM & ΔZLM

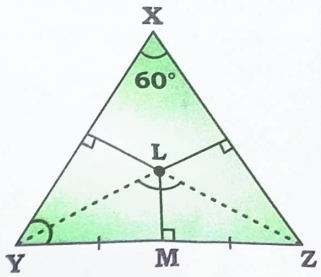
YM = MZ

$\angle YML = \angle LMZ = 90°$

LY = LZ (Circum radius)

$\Rightarrow \Delta YLM \cong \Delta ZLM$ (By RHS)

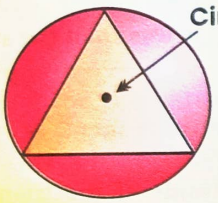
$\therefore \angle YLM = \angle ZLM = \frac{120°}{2} = 60°$



Position of circum-centre

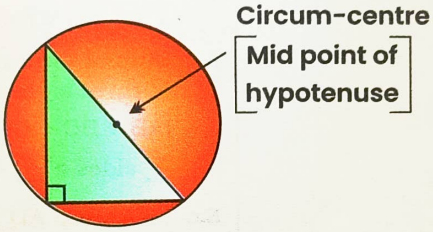
Acute angle triangle

In an acute angle triangle, the circum-centre is always inside the triangle.



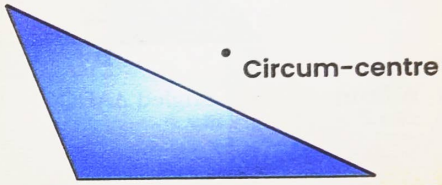
Right angle triangle

In a right-angled triangle, the circum-centre is at the mid-point of the hypotenuse.



Obtuse angle triangle

In an obtuse-angled triangle, the circum-centre is always outside the triangle and will be at the front of the angle which is obtuse.



Circum-radius

Circum-radius of any triangle

$$R = \frac{\text{Product of all three sides}}{4 \times \text{Area of the triangle}} = \frac{abc}{4\Delta}$$

Circum-radius of right angle triangle

$$R = \frac{\text{Hypotenuse}}{2} = \frac{h}{2}$$

Circum-radius of equilateral triangle

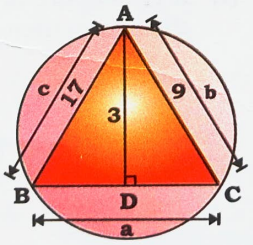
$$R = \frac{\text{Side}}{\sqrt{3}} = \frac{a}{\sqrt{3}}$$

Ex. In a ΔABC, AB = 17 cm, AC = 9 cm, AD is perpendicular on BC & AD = 3 cm. Find the circum radius of this triangle.

HINTS

$$R = \frac{abc}{4\Delta} = \frac{a \times 9 \times 17}{4 \times \frac{1}{2} \times a \times 3}$$

= 25.5 cm

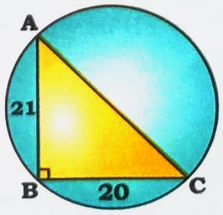


Ex. The lengths of the two sides forming the right angle of a right-angled triangle are 21 cm and 20 cm. What is the radius of the circle circumscribing the triangle?

HINTS

AC = 29 (By triplet of 20, 21, 29)

$$R = \frac{\text{Hypotenous}}{2} = \frac{29}{2} = 14.5 \text{ cm}$$



Ex. ΔABC is an equilateral triangle. If the area of the triangle is $36\sqrt{3}$ then what is the radius of circle circumscribing the ΔABC?

HINTS $\frac{\sqrt{3}}{4}a^2 = 36\sqrt{3}$

$\Rightarrow a^2 = 144$

$\Rightarrow a = 12$

$\therefore R = \frac{a}{\sqrt{3}} = \frac{12 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$

$= 4\sqrt{3}$ cm

Relationship between semi-perimeter, inradius & circum-radius of right angle triangle

- Area of Δ = rs

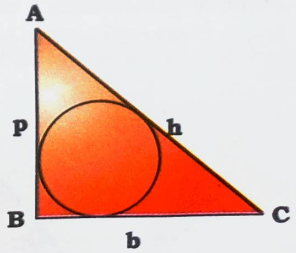
  Put, r = s – 2R

  $\Delta = s(s - 2R)$

- Δ = rs

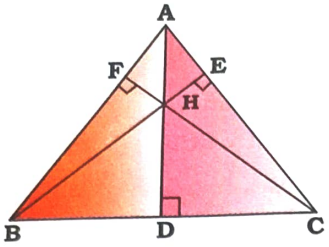
  Put, s = r + 2R

  $\Delta = r(r + 2R) = r^2 + 2Rr$



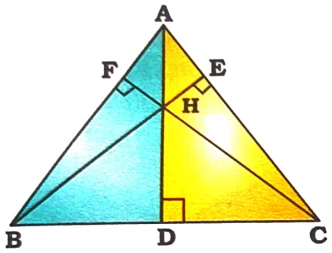
Orthocentre

Altitude: If a perpendicular is drawn from a vertex of a triangle to the opposite side is called the altitude of the triangle.

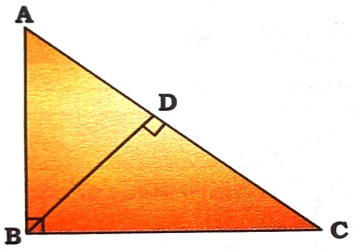


If AD ⊥ BC, therefore AD is the altitude.

(i) Altitudes in acute angle ΔABC

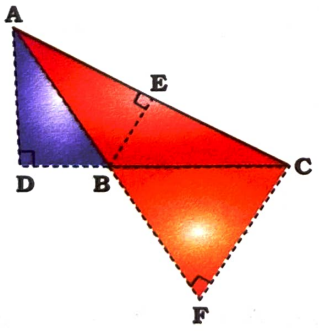


(ii) Altitude in right-angled ΔABC



Two altitudes of a right-angled triangle ABC are AB and BC and the third altitude is BD.

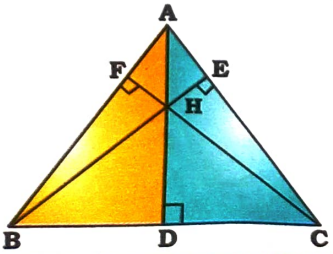
(iii) Altitude in obtuse-angled ΔABC



Three altitudes of ΔABC are AD, BE and CF.

Orthocentre

Point of intersections of the altitudes of a triangle is called the orthocentre of the triangle. It is generally denoted by H.



Properties of Orthocentre

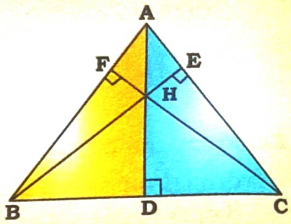
(i) All the three altitudes of a triangle pass through a point (orthocentre)

(ii) $\angle BHC = 180° - \angle A$

$\angle CHA = 180° - \angle B$

$\angle AHB = 180° - \angle C$

It is true for all types of triangle.



Ex. In an obtuse-angled triangle ABC, ∠A is the obtuse angle and O is the orthocentre. If ∠BOC = 54°, then ∠BAC is:

HINTS

$\angle BAC = 180° - 54° = 126°$

(iii) ΔBFH ~ ΔCEH

$BH \times HE = CH \times HF$

Also, $BH \times HE = CH \times HF = AH \times HD$

(iv) Sum of three altitudes of a triangle is less than the sum of three sides of triangle.

$(AB + BC + CA) > (AD + CF + BE)$

$\sum$ sides > $\sum$ Altitudes

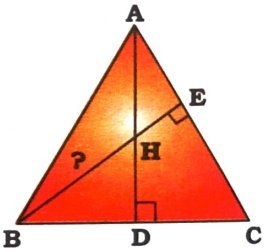
Ex. In ΔABC, AD and BE are the altitude of the triangle meets at H such that AH = 12 cm, HD = 9 cm, and HE = 4 cm. Find BH?

HINTS

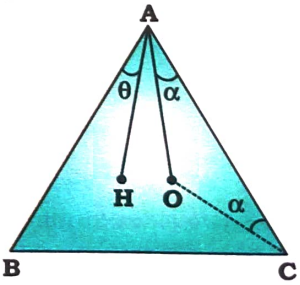
$AH \times HD = BH \times HE$

$\Rightarrow 12 \times 9 = BH \times 4$

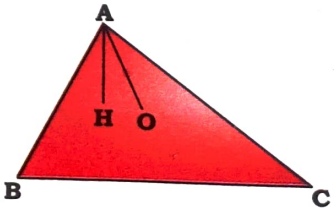
$\Rightarrow BH = 27$ cm



(v) If H & O are orthocentre & circumcentre (as shown in the figure) then, θ = α



Ex. In ΔABC, H is the orthocenter, O is circumcenter. If ∠BAH = 30° then find ∠OAC?



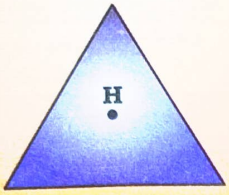
HINTS Given, ∠BAH = 30°

∠OAC = ∠BAH = 30°

Position of orthocentre

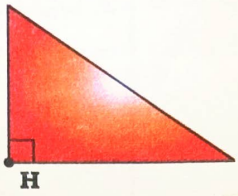
Acute angle triangle

In an acute-angled triangle the orthocentre is always inside the triangle.



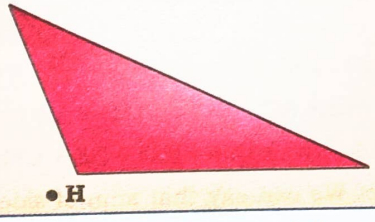
Right angle triangle

In right-angled triangle the orthocentre is on the vertex of the triangle at which triangle is the right-angled.



Obtuse angle triangle

In an obtuse-angled triangle the orthocentre is always outside of the triangle and will be at the back of the angle which is obtuse.





- Distance between orthocentre and circum-centre of a right angle triangle $= \frac{\text{Hypotenuse}}{2} = \frac{h}{2}$
- Distance between circum-centre and incentre in any triangle $= \sqrt{R^2 - 2Rr}$

Where, R = circum-radius & r = inradius

Ex. If the circumradius of a triangle is 6 cm and inradius is 2 cm. Find the distance between circum-centre and incentre.

HINTS R = 6, r = 2

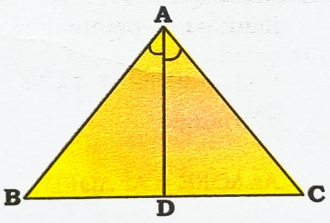
$\therefore d = \sqrt{R^2 - 2Rr}$

$= \sqrt{36 - 2\times6\times2} = \sqrt{12} = 2\sqrt{3}$

Centroid

Median: If a line segment from a vertex of a triangle bisects the opposite side, the line segment is called the median of the triangle.

If line segment AD bisects BC, then AD is the median.



Centroid: The point of intersection of the medians of the triangle is called the centroid. It is denoted by G.

Properties of Centroid

(i) All the three medians of a triangle pass through a point (Centroid).

(ii) Centroid of all types of triangles are always inside the triangle.

(iii) Apollonius Theorem

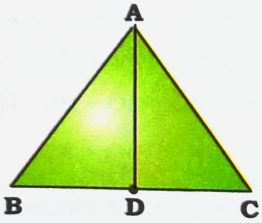
If AD is median of ΔABC, then

$AB^2 + AC^2 = 2(AD^2 + CD^2)$

'OR'

Length of median (AD)

$= \frac{1}{2}\sqrt{2AC^2 + 2AB^2 - BC^2}$



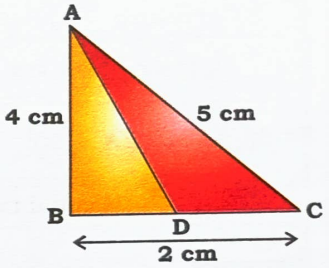
Ex. If the sides of ΔABC are 5 cm, 4 cm and 2 cm, then find the length of the median inserted from its smallest angled vertex?

HINTS We know that smallest angle is the angle opposite to smallest side.

$$AD = \frac{\sqrt{2AB^2 + 2AC^2 - BC^2}}{2}$$

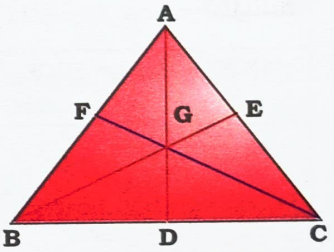
$$= \frac{\sqrt{32 + 50 - 4}}{2}$$

$$= \frac{\sqrt{78}}{2} = \sqrt{\frac{39}{2}} \text{ cm}$$



(iv) If AD, BE and CF are medians of ΔABC and G is the centroid, then

$$\frac{AG}{GD} = \frac{BG}{GE} = \frac{CG}{GF} = \frac{2}{1}$$



(v) $\frac{AG^2 + BG^2 + CG^2}{AB^2 + BC^2 + CA^2} = \frac{1}{3}$

Ex. G is the centroid of ΔABC. If the length of sides of ΔABC are 8 cm, 10 cm, and 12 cm, then find the $AG^2 + BG^2 + CG^2$?

HINTS

$$AG^2 + BG^2 + CG^2 = \frac{1}{3}[64 + 100 + 144] = \frac{308}{3}$$

(vi) $3(AB^2 + BC^2 + CA^2) = 4(AD^2 + BE^2 + CF^2)$

Ex. In ΔABC, the sum of square of all sides is 64 cm. Find the sum of square of all three medians.

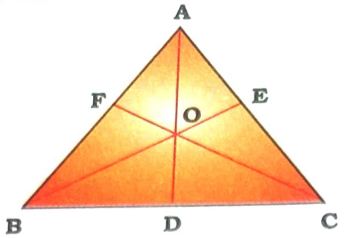
HINTS

$3(a^2 + b^2 + c^2) = 4(m_a^2 + m_b^2 + m_c^2)$

$\Rightarrow 3 \times 64 = 4 \times R$

$\Rightarrow R = 48$

(vii) The sum of any two sides of a triangle is greater than twice the median drawn to the third side.



$AB + AC > 2AD$ .....(i)

$AB + BC > 2BE$ .....(ii)

$AC + BC > 2CF$ .....(iii)

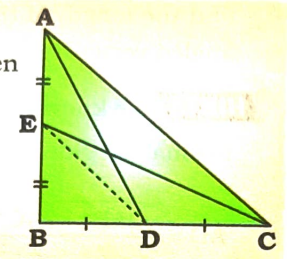
Adding (i), (ii) and (iii)

$2(AB + AC + BC) > 2(AD + AE + CF)$

$(AB + AC + BC) > (AD + BE + CF)$

So, We can say that sum of sides (perimeter) is always greater than sum of all median.

(viii) If ΔABC is Right angle triangle and AD & CE are medians, then



- $AC^2 = 4ED^2$
- $4(AD^2 + CE^2) = 5AC^2$

$\because AD^2 + CE^2 = AC^2 + ED^2$

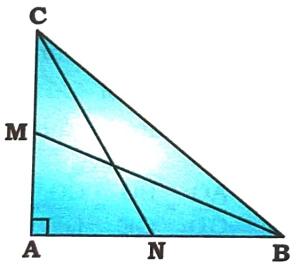
$\therefore AD^2 + CE^2 = 5ED^2$

Ex. In ΔABC, $\angle A = 90°$, if BM and CN are two medians, $\frac{BM^2 + CN^2}{BC^2}$ is equal to:

HINTS

$4(CN^2 + BM^2) = 5BC^2$

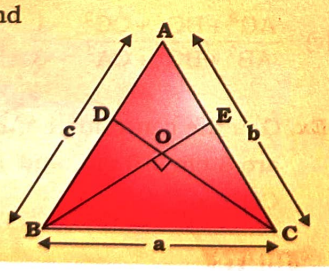
$\Rightarrow \frac{BM^2 + CN^2}{BC^2} = \frac{5}{4}$



(ix) If BE & CD are medians and $BE \perp CD$, then

$AB^2 + AC^2 = 5BC^2$

$\Rightarrow c^2 + b^2 = 5a^2$



Ex. If In ΔABC, BE and CF are two medians perpendicular to each other and if AB = 19 cm and AC = 22 cm then the length of BC is:

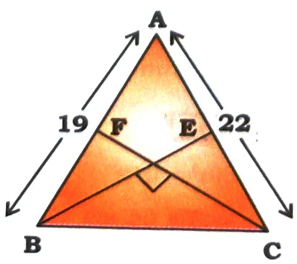
HINTS $AB^2 + AC^2 = 5BC^2$

$\Rightarrow 19^2 + 22^2 = 5BC2$

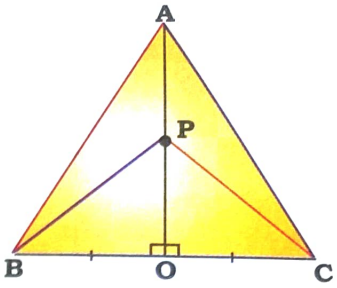
$\Rightarrow 361 + 484 = 5BC^2$

$\Rightarrow 845 = 5BC^2 \Rightarrow BC^2 = 169$

$\Rightarrow BC = 13$

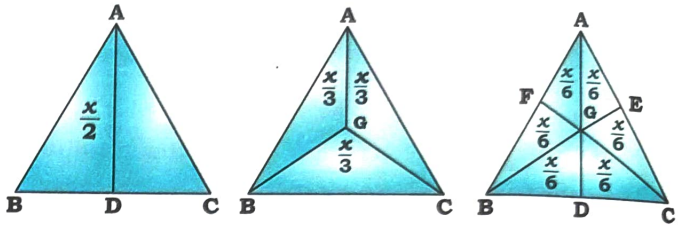


(x)



$AB^2 + PC^2 = AC^2 + BP^2$

(xi) If the area of $\Delta ABC = x$, AD, BE and CF are three medians and G is centroid, then

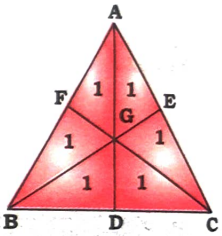


Ex. In a ΔABC, AD, BE and CF are medians intersect at G. If area of ΔABC is 204 cm^2. Find the area of quadrilateral GDCE.

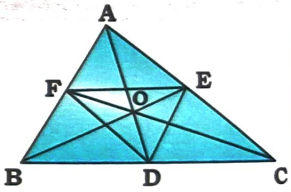
HINTS

Here, 6 unit = 204 cm^2

$\therefore$ 2 unit = 68 cm^2



(xii) Area of triangle formed by joining mid points of two sides and centroid is $\frac{1}{12}$th of area of triangle.



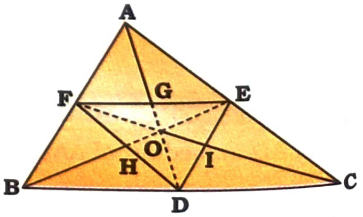
Ar ΔOFE = Ar ΔOFD

= Ar $\Delta OED = \frac{1}{12}$ Ar ΔABC

$\Delta OFE \sim \Delta OCB$

(According to Mid-Point theorem FE || BC so that $\angle EFO = \angle OCB$, $\angle OEF = \angle OBC$)

(xiii) The line segment joining the mid points of two sides divides the line joining of vertex in between line to the centroid in the ratio 3 : 1.



AG : GO = BH : HO

= CI : IO = 3 : 1

G is the Mid point of AD and FE

(xiv) If three medians of a triangle are given then

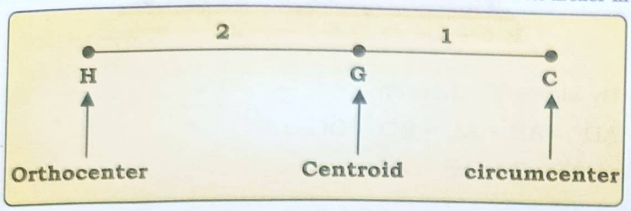
Area of $\Delta = \frac{4}{3}$ (Area of triangle considering medians as side)

$= \frac{4}{3}\sqrt{S_m(S_m - m_1)(S_m - m_2)(S_m - m_3)}$

Where, $S_m = \frac{m_1 + m_2 + m_3}{2}$

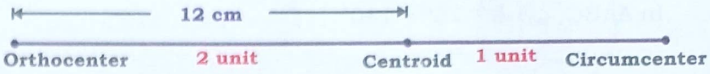
Relation between Orthocenter, centroid and circumcenter

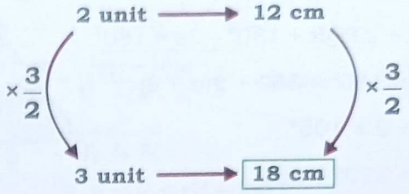
In any triangle, orthocenter, centroid and circumcenter are co-linear and centroid divides a line in 2:1. This line is also known as Euler line.



Ex. In a triangle the distance between the centroid and orthocentre is 12 cm. Find the distance between orthocentre and circumcenter.

HINTS





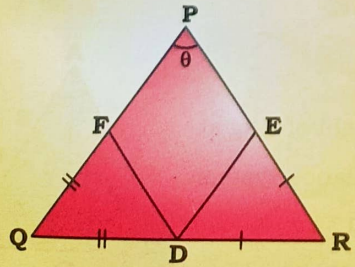
∴ Distance between orthocentre & circumcenter = 18 cm

Summary

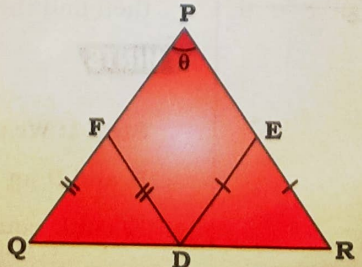
Feature	Incenter	Circumcenter	Orthocenter	Centroid
Symbol	I	O	H	G
Definition	Intersection point of internal angle bisectors.	Intersection of perpendicular bi-sectors of sides.	Intersection of altitudes	Intersection of medians (line segment from vertex to midpoint of opposite side)
Associated circle	Centre of the inscribed circle (incircle)	Centre of circumscribed circle (circumcircle)	No direct associated circle as a center	No direct associated circle as a center
Equidistant from	Sides of the triangle	Vertices of the triangle	Not equidistant from sides or vertices	Not equidistant from sides or vertices
Location	Always inside the triangle	• Acute: Inside • Right: Midpoint of hypotenuse • Obtuse: Outside at the front of obtuse angle	• Acute: Inside • Right: At the right angle vertex • Obtuse: Outside at the back of obtuse angle	Always inside the triangle
Significance	Center of largest circle that fits inside	Center of circle passing through all vertices	Related to triangle's heights	Center of mass/gravity of the triangle

Some important results

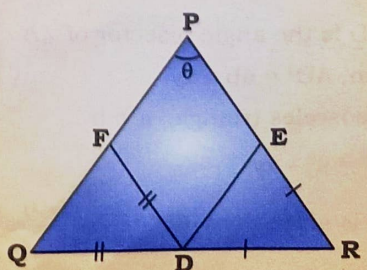
In ΔPQR, D, E, F are points on sides QR, PR and PQ respectively. If QF = QD, RE = DR and ∠RPQ = θ then $\angle EDF = 90° - \frac{\theta}{2}$



In ΔPQR, D, E, F are points on sides QR, PR and PQ respectively. If QF = FD, DE = ER and ∠RPQ = θ then ∠EDF = θ



In ΔPQR, D, E, F are points on sides QR, PR and PQ respectively. If QD = DF, DR = RE and ΔRPQ = θ then ∠EDF = 180° – 2θ



Ex. In a ΔABC, points P, Q and R are taken on AB, BC and CA, respectively, such that BQ = PQ and QC = QR. If ∠BAC = 75°, what is the measure of ∠PQR (in degrees)?

HINTS

In ΔABC, $\alpha + \theta + 75° = 180°$

$\Rightarrow \alpha + \theta = 105°$

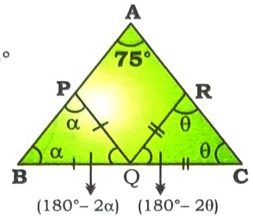
$\because$ BQC is a straight line.

$180° - 2\alpha + \angle PQR + 180° - 2\theta = 180°$

$\Rightarrow \angle PQR = 180° - 360 + 2(\alpha + \theta)$

$= -180° + 2 \times 105°$

$= -180° + 210° = 30°$



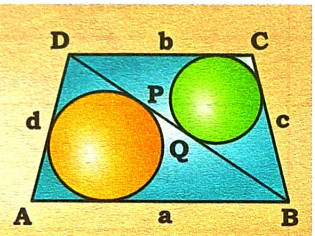
Alternatively

$\angle PQR = 180° - 2 \times \angle BAC$

$= 180° - 2 \times 75° = 180° - 150° = 30°$

☞ In trapezium ABCD, two unequal circles are inscribed such that each circle touches one pair of non-parallel sides. The diagonal DB intersects the circles at point P & Q.

$PQ = \dfrac{(a+b)-(c+d)}{2}$



☞ If n equal circles fit perfectly along base BC of ΔABC, each touching adjacent circles and the Triangle's sides. then their radius is.

$r = \dfrac{\text{Area of } \Delta ABC}{s + (n-1) \times h}$

where, s = semi-perimeter

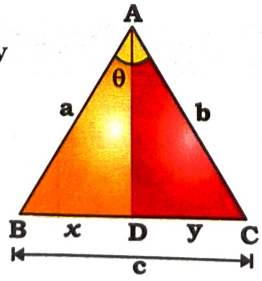
Some important theorems

(A) Stewart theorem:

It is used to find the length of any cevian.

Where $c = x + y$

$a^2y + b^2x = AD^2c + xyc \Rightarrow a^2y + b^2x = c(AD^2 + xy)$



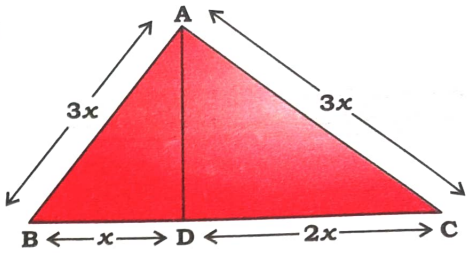
- If AD is the angle bisector of ∠A

  Then, $AD^2 = ab - xy$
- In isosceles triangle, a = b

  $AD^2 = a^2 - xy$
- If θ is given then $AD = \dfrac{2ab\cos\theta}{a+b}$

Ex. ΔABC is an equilateral triangle. D is a point on side BC such that BD : BC = 1 : 3. If $AD = 5\sqrt{7}$ cm, then the side of the triangle is:

HINTS



By stewart's theorem,

$AD^2 = AB \times AC - BD \times DC$

$\Rightarrow 25 \times 7 = 9x^2 - 2x^2$

$\Rightarrow 7x^2 = 25 \times 7$

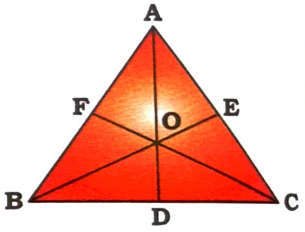
$\Rightarrow x = 5$

$\therefore$ Side $= 3x = 3 \times 5 = 15$ cm

(B) Ceva Theorem

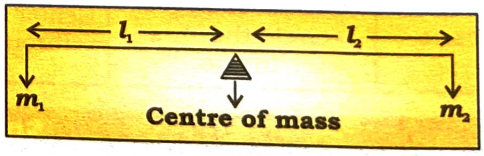
In ΔABC, AD, BE and CF are the cevians which pass through a common point O.

- $\dfrac{AF}{FB} \times \dfrac{BD}{DC} \times \dfrac{CE}{EA} = 1$
- $\dfrac{OD}{AD} + \dfrac{OE}{BE} + \dfrac{OF}{CF} = 1$
- $\dfrac{AO}{AD} + \dfrac{BO}{BE} + \dfrac{CO}{CF} = 2$



(C) Mass Point Theorem

- It is used to solve problems involving triangles and intersecting cevians by applying centers of mass principles.
- The base idea is that of a see-saw with weights at each end. The see-saw will balance if the product of mass and its distance to the fulcrum is same for each mass.

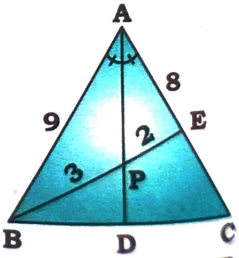


$m_1 l_1 = m_2 l_2 \Rightarrow \dfrac{m_1}{m_2} = \dfrac{l_2}{l_1}$

Ex. In ΔABC, AD is angle bisector of ∠A. AD and BE lie on BC and AC respectively such that AB = 9 cm and AC = 8 cm. AD and BE intersects at point P and BP : PE = 3 : 2, then find the value of AE.

HINTS

Step 1: We put the weight of 8 kg at B and 9 kg at C to balance system BC at D equivalent to 17 kg.



Step 2: To balance AC, we put 3 kg at A and getting resultant at E equivalent to 12 kg.

The ratio of the distances from the centre of mass would be inverse of the weights at the end points.

Therefore, AE : EC = 9 : 3 = 3 : 1

$\therefore$ (3 + 1) = 4 unit $\xrightarrow{\times 2}$ 8 cm

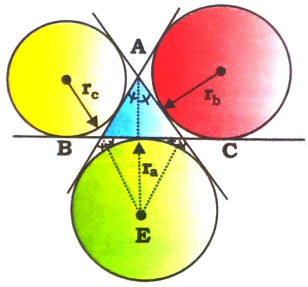
1 unit $\xrightarrow{\times 2}$ 2 cm

$\therefore$ AE = 3 unit $\xrightarrow{\times 2}$ 6 cm

AE = 6 cm

Excenter

The intersection point of internal angle bisector of one angle and bisectors of other two opposite exterior angles.



(a) $\angle BEC = 90° - \frac{\angle A}{2}$

(b) Ex-radii

$$r_a = \frac{\Delta}{s-a}; r_b = \frac{\Delta}{s-b}; r_c = \frac{\Delta}{s-c}$$

(c) $r_a = \frac{rs}{s-a} = \sqrt{\frac{s(s-b)(s-c)}{s-a}}$ where, $s = \frac{a+b+c}{2}$

(d) $\Delta(\text{area}) = \sqrt{r\, r_a\, r_b\, r_c}$

(e) $r^2_a + r^2_b + r^2_c = (4R + r)^2 - 2s^2$